

Symptom: Ping shows consistently high latency to a destination, but no packet loss

Overview

Round-trip time is elevated across every attempt (not spiky or intermittent), while every packet still eventually arrives. Consistent high latency without loss usually points to a genuine path/distance or processing-delay issue, rather than congestion or instability, which tend to show as loss or jitter instead.

Cause 1: Suboptimal Routing Path

Traffic may be taking a longer physical or logical path than necessary — for example, routed through a distant data center due to a misconfigured route preference, or via an inefficient path caused by asymmetric routing. Every packet still gets delivered correctly, but the extra hops or distance adds consistent delay.

Diagnostic steps:

- Run a traceroute to the destination and review the hop-by-hop path and per-hop latency to identify where delay is being introduced
- Compare the current path against a known-good or expected path (if the topology is documented) to spot an unexpected detour
- Check routing table entries and route preference/metric configuration for anything that could be steering traffic through a longer-than-necessary path

Cause 2: High Processing Delay at an Intermediate Device

A router or firewall along the path that's under sustained CPU load may still forward every packet correctly, but with added per-packet processing delay, contributing consistent latency without causing any actual drops.

Diagnostic steps:

- Correlate traceroute hop latency against the CPU utilization of any manageable devices along the path
- Test during a lower-traffic period to see if latency improves, which would support a load-related processing delay rather than a fixed path/distance issue

Symptom: Ping or traceroute shows intermittent packet loss, with latency that spikes unpredictably (high jitter)

Overview

Unlike consistent high latency, this pattern — some packets fast, others slow or lost entirely, no stable pattern — points toward congestion, queueing, or an unstable link rather than a fixed routing/distance issue.

Cause 1: Interface or Link Congestion

A link operating near or at its bandwidth capacity will start queueing packets during traffic bursts, and once the queue fills, further packets are dropped. This produces exactly the "sometimes fast, sometimes slow, sometimes lost" pattern, correlating with traffic volume rather than being constant.

Diagnostic steps:

- Check interface utilization on links along the path during the period of observed loss/jitter
- Look for output queue drops or discard counters on relevant interfaces, which directly indicate congestion-driven loss
- Correlate the timing of reported issues against known high-traffic periods (backups, batch jobs, peak business hours)

Cause 2: Physical Layer Errors (Bad Cable, Duplex Mismatch)

A failing cable, a marginal SFP/transceiver, or a duplex/speed mismatch between two connected interfaces can cause intermittent packet corruption and loss that has nothing to do with traffic volume or routing — it's a physical-layer problem masquerading as a network performance issue.

Diagnostic steps:

- Check interface error counters (CRC errors, collisions, runs) on both ends of each link along the path — a healthy link should show these near zero
- Confirm speed/duplex settings match on both ends of each connection; a mismatch (one side auto-negotiating, the other hard-set) is a classic, easy-to-overlook cause
- Physically inspect or swap the cable/transceiver as a quick test if error counters point to a specific link

Symptom: Traceroute shows the destination is unreachable, with the path stopping (timing out) at a specific hop

Overview

Unlike general slowness or loss, this is a hard failure at a specific point in the path — everything before that hop responds normally, and nothing beyond it does. Localizing exactly where the path breaks is most of the diagnostic work here.

Cause 1: Firewall or ACL Blocking Traffic at That Hop

A device along the path — often a firewall, sometimes a router with an access list — may be configured to silently drop the traceroute's probe packets (commonly UDP or ICMP, depending on the tool) without sending back the expected response, making it appear as though the path stops there, even if actual application traffic might still get through via a different protocol/port.

Diagnostic steps:

- Test connectivity to the same destination using the actual application protocol/port in question (not just ICMP-based traceroute), since a firewall may block traceroute probes specifically while still allowing the real traffic
- Check firewall/ACL logs on the device at or immediately after the last-responding hop for explicit deny entries matching the traceroute's traffic pattern
- Try an alternate traceroute method (TCP-based instead of UDP/ICMP-based) if available, since some paths block one probe type but not another

Cause 2: Routing Loop or Missing Route at That Hop

If a device along the path has no route to the destination (a missing or recently withdrawn route) or is caught in a routing loop (traffic bouncing between two devices, each believing the other has the correct path), traceroute will show the path stalling or looping around that point rather than progressing.

Diagnostic steps:

- Check the routing table on the device at the last-responding hop for a valid route to the destination network
- If a loop is suspected (the same one or two hops repeating in the traceroute output), check both suspected devices' routing tables for a circular route preference

Symptom: Performance is fine for most destinations, but consistently poor to one specific remote site or subnet

Overview

General network health looks normal — most traffic performs well — but one particular destination or site is reliably slow or lossy. Isolating the problem to a single destination narrows the cause to something specific to that path or site, rather than a general infrastructure issue.

Cause 1: Undersized or Saturated WAN Link to That Specific Site

If the affected destination is reached via a dedicated WAN link (e.g., a branch office circuit) that's simply undersized for current traffic demand, or has degraded (a partially failing circuit still passing some traffic), performance to that specific site will suffer while traffic to everywhere else — routed via different, healthy links — remains fine.

Diagnostic steps:

- Check utilization and error rates specifically on the WAN circuit serving the affected site
- Compare current traffic volume to that site against the circuit's provisioned capacity
- Contact the circuit provider for a line-quality/error check if utilization alone doesn't explain the degradation

Cause 2: Asymmetric or Suboptimal Route Specific to That Destination

A routing policy, static route, or BGP preference that was correctly configured at one point may no longer reflect the best path if the network topology has changed (a link added, a provider changed, a site migrated) — causing traffic to that one destination specifically to take a longer or lower-quality path while other destinations are unaffected.

Diagnostic steps:

- Traceroute to the affected destination and compare the path against traceroutes to other, unaffected destinations to spot anything unusual specific to this route
- Review any static routes or routing policy specific to that destination's network/subnet for outdated configuration